

High-Dimensional Panel Data Models with Interactive Fixed Effects: Beyond the Linear Case

Maximilian Rücker
Ulm University



universität
uulm

Joint work with Michael Vogt (Ulm University) and Oliver Linton (University of Cambridge)

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Michael Vogt (Ulm University)



Oliver Linton (University of Cambridge)

Model setting

- Observe panel data

$$\{(Y_{it}, X_{it}) : 1 \leq i \leq n, 1 \leq t \leq T\}, \quad X_{it} \in \mathbb{R}^p,$$

where p may be larger than nT .

- **Linear model** studied in Rücker et al. (2025):

$$Y_{it} = \sum_{j=1}^p X_{it,j} \beta_j + F_t^\top \gamma_i + \varepsilon_{it}.$$

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- **Linear model** studied in Rücker et al. (2025):

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- **Parametric additive model** studied here:

$$Y_{it} = \sum_{j=1}^p m_j(X_{it,j}) + F_t^\top \gamma_i + \varepsilon_{it}, \quad m_j(x) = \sum_{\ell=1}^{L_j} \phi_\ell(x) \beta_{j\ell}.$$

Example: $L_j = 3$, $\phi_\ell(x) = x^\ell$, $m_j(x) = x\beta_{j1} + x^2\beta_{j2} + x^3\beta_{j3}$.

Aim: Develop estimation procedure for m_j , that is the coefficients $\beta_{j\ell}$.

Interactive fixed effects

- ▶ Error component:

$$e_{it} = F_t^\top \gamma_i + \varepsilon_{it}.$$

- ▶ Covariate structure, Pesaran (2006):

$$X_{it} = \Gamma_i F_t + Z_{it}.$$

- ▶ The loadings γ_i and Γ_i may be correlated.
- ▶ This creates **factor-induced endogeneity**: regressors and errors share the same latent factors.

Interactive fixed effects generalize two-way fixed effects.

Why the linear case is simpler

Stack observations over time:

$$Y_i = \mathbf{X}_i\beta + \mathbf{F}\gamma_i + \varepsilon_i, \quad \mathbf{X}_i = \mathbf{F}\boldsymbol{\Gamma}_i^\top + \mathbf{Z}_i,$$

for $1 \leq i \leq n$, where

$$\underset{(T \times 1)}{Y_i} = \begin{pmatrix} Y_{i1} \\ \vdots \\ Y_{iT} \end{pmatrix}, \quad \underset{(T \times p)}{\mathbf{X}_i} = \begin{pmatrix} X_{i1}^\top \\ \vdots \\ X_{iT}^\top \end{pmatrix}, \quad \underset{(T \times K)}{\mathbf{F}} = \begin{pmatrix} F_1^\top \\ \vdots \\ F_T^\top \end{pmatrix}, \dots$$

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Let $\mathbf{\Pi}$ be the projection onto the orthogonal complement of the factor space $\mathbf{F}$. Then

$$\mathbf{\Pi} Y_i = \mathbf{\Pi} \mathbf{X}_i \beta + \mathbf{\Pi} \varepsilon_i, \quad \mathbf{\Pi} \mathbf{X}_i = \mathbf{\Pi} \mathbf{Z}_i.$$

In the **linear** model, projection removes the factor component from both the error and the design.

What changes beyond the linear case?

In the additive model

$$\begin{aligned} Y_{it} &= \sum_{j=1}^p m_j(X_{it,j}) + F_t^\top \gamma_i + \varepsilon_{it} \\ &= \sum_{j=1}^p \sum_{\ell=1}^{L_j} \phi_\ell(X_{it,j}) \beta_{j\ell} + F_t^\top \gamma_i + \varepsilon_{it} \end{aligned}$$

the transformed regressors are

$$\phi_\ell(X_{it,j}) = \phi_\ell(\Gamma_{i,j}^\top F_t + Z_{it,j})$$

with coefficients $\beta_{j\ell}$.

What changes beyond the linear case?

Stacking the equations across t yields

$$Y_i = \Phi_i \beta + \mathbf{F} \gamma_i + \varepsilon_i,$$

for $1 \leq i \leq n$ where

$$\Phi_i = (\Phi_{i,1}, \dots, \Phi_{i,p}) \in \mathbb{R}^{T \times \sum_{j=1}^p L_j}$$

$$\Phi_{i,j} = (\phi_1(X_{i \cdot j}), \dots, \phi_{L_j}(X_{i \cdot j})) \in \mathbb{R}^{T \times L_j}.$$

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After projection

$$\Pi Y_i = \Pi \Phi_i \beta + \Pi \varepsilon_i,$$

the design $\Pi \Phi_i = \Pi \Phi (\Gamma_i F^\top + Z_i)$ still contains the factors.

Endogeneity issue is solved; central problem: verifying that the projected nonlinear design is **regular** enough.

Estimation strategy

How to estimate the unknown components $\beta_{j\ell}$?

- **Step 1:** Stack equations across t

$$Y_i = \Phi_i \beta + F \gamma_i + \varepsilon_i,$$

- **Step 2:** Center the equations

$$\tilde{Y}_i = \tilde{\Phi}_i \beta + F \tilde{\gamma}_i + \tilde{\varepsilon}_i,$$

where $\tilde{Y}_i = Y_i - \frac{1}{n} \sum_{i=1}^n Y_i, \dots$ denote the centered quantities.

- **Step 3:** Project away the unknown factors F with some suitable projection matrix $\hat{\Pi}$:

$$\hat{\Pi} \tilde{Y}_i = \hat{\Pi} \tilde{\Phi}_i \beta + \underbrace{\hat{\Pi} F \tilde{\gamma}_i}_{\approx 0} + \hat{\Pi} \tilde{\varepsilon}_i$$

Estimation strategy

- **Step 4:** Run lasso on the projected data $\{(\hat{\Pi}\tilde{Y}_i, \hat{\Pi}\tilde{\Phi}_i), i = 1, \dots, n\}$:

$$\hat{\beta}_{\lambda} \in \arg \min_{b \in \mathbb{R}^{\sum_j L_j}} \left\{ \frac{1}{nT} \sum_{i=1}^n \left\| \hat{\Pi}\tilde{Y}_i - \hat{\Pi} \sum_{j=1}^p \tilde{\Phi}_{i,j} b_j \right\|_2^2 + \lambda \|b\|_1 \right\}.$$

- We call $\hat{\beta}_{\lambda}$ the **HD-CCE estimator**.

Construction of $\hat{\Pi}$

- ▶ Pesaran's CCE idea: use cross-sectional averages to recover the factor space.

$$\bar{\mathbf{X}} = \frac{1}{n} \sum_{i=1}^n \mathbf{X}_i \approx \mathbf{F} \mathbb{E}[\boldsymbol{\Gamma}_i]^\top.$$

- ▶ Classical CCE projection fails when $p \geq T$, because $\bar{\mathbf{X}}$ likely spans all of $\mathbb{R}^T$.
- ▶ We estimate the number of factors and retain only the leading spectral directions:

$$\bar{\mathbf{W}} = \bar{\mathbf{X}} \hat{\mathbf{U}}_{1:\hat{K}}, \quad \hat{\Pi} = \mathbf{I} - \bar{\mathbf{W}}(\bar{\mathbf{W}}^\top \bar{\mathbf{W}})^{-1} \bar{\mathbf{W}}^\top.$$

Compatibility condition

The projected nonlinear design satisfies a compatibility condition if

$$\left(\sum_{(j,\ell) \in S} |b_{j\ell}| \right)^2 \leq \frac{|S|}{\varphi^2} \frac{1}{nT} \sum_{i=1}^n \left\| \hat{\Pi} \tilde{\Phi}_i b \right\|_2^2$$

for all b in the cone

$$\sum_{(j,\ell) \in S^c} |b_{j\ell}| \leq 3 \sum_{(j,\ell) \in S} |b_{j\ell}|.$$

This is the high-dimensional eigenvalue condition that makes lasso consistent.

Such a condition is close to necessary, see e.g. Bickel et al. (2009).

Theorem 4.4 [Rücker et al. (2026)]

Suppose that the projected nonlinear design satisfies a compatibility condition. Then under certain regularity conditions

$$\|\hat{\beta}_\lambda - \beta\|_1 = O_p(s\lambda), \quad s = |S|$$

where

$$\lambda = h_n \sqrt{\frac{\log(pT)}{n}} (npT)^{2/\theta}.$$

for some slowly diverging sequence h_n and θ denotes the model moments.

► The rate

$$s \cdot \lambda = \frac{1}{\sqrt{n}} \cdot s \cdot h_n \sqrt{\log(pT)} (npT)^{2/\theta}$$

is driven mainly by n , not by nT , because the theory allows strong dependence over time (cf. Hansen (2007)).

Verification of Compatibility condition

- Step 1 (Theorem 4.7): Reduce sample compatibility condition for the empirical matrix

$$\frac{1}{nT} \sum_{i=1}^n \tilde{\boldsymbol{\Phi}}_i^\top \hat{\boldsymbol{\Pi}} \tilde{\boldsymbol{\Phi}}_i$$

to the population compatibility condition for

$$\frac{1}{T} \mathbb{E} \left[(\boldsymbol{\Phi}_1 - \mathbb{E}[\boldsymbol{\Phi}_1])^\top \boldsymbol{\Pi} (\boldsymbol{\Phi}_1 - \mathbb{E}[\boldsymbol{\Phi}_1]) \right].$$

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- Step 2 (Theorem 4.8): Under Gaussian regressors and polynomial dictionaries, verify the population condition via Hermite expansions.

Monte-Carlo experiments: Simulation design (I)

- ▶ Cross-section fixed at $n = 50$; two time dimensions:
 $T \in \{15, 50\}$. Number of covariates $p = 600$.
- ▶ Factors $F_{t,k} = 0.5F_{t-1,k} + w_{t,k}$, $w_{t,k} \sim N(0, 0.75)$.
- ▶ Regressors

$$X_{it} = \Gamma_i F_t + Z_{it}$$

with idiosyncratic part $Z_{it} \in \mathbb{R}^{3d} \sim N(0, I)$ and loadings

$$\Gamma_i = \begin{pmatrix} \Gamma_i^{(1)} & 0 & 0 \\ 0 & \Gamma_i^{(2)} & 0 \\ 0 & 0 & \Gamma_i^{(3)} \end{pmatrix} \in \mathbb{R}^{3d \times 3}$$

with random vectors $\Gamma_i^{(1)} = (\Gamma_{i,1}, \dots, \Gamma_{i,d})^\top$,
 $\Gamma_i^{(2)} = (\Gamma_{i,d+1}, \dots, \Gamma_{i,2d})^\top$ and $\Gamma_i^{(3)} = (\Gamma_{i,2d+1}, \dots, \Gamma_{i,3d})^\top$ of
 length $d = p/3$.

Simulation design (II)

- $G_i = (\gamma_i^\top, \{\Gamma_i^{(1)}\}^\top, \{\Gamma_i^{(2)}\}^\top, \{\Gamma_i^{(3)}\}^\top)^\top \sim N(\mu, \Omega)$
independent. μ is a vector of ones and

$$\Omega = \begin{pmatrix} 1 & \rho & \cdots & \rho \\ \rho & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \rho \\ \rho & \cdots & \rho & 1 \end{pmatrix},$$

with $\rho = 0.75$.

Simulation design (III)

- ▶ The response is generated according to

$$Y_{it} = \sum_{j=1}^p m_j(X_{it,j}) + F_t^\top \gamma_i + \varepsilon_{it}, \quad m_j(x) = \sum_{\ell=1}^{L_j} x^\ell \beta_{j\ell}, \quad L_j = 5.$$

- ▶ ε_{it} is iid $N(0, 1)$.
- ▶ $\beta_{j\ell}$ are set such that

$$m_1(x) = 0.2x + 0.1x^2 + 0.01x^3,$$

$$m_2(x) = 0.2x,$$

$$m_3(x) = 0.1x^2 + 0.01x^3,$$

$$m_4(x) = 0.1x^2,$$

$$m_5(x) = 0.2x + 0.01x^3,$$

$$m_6(x) = 0.2x + 0.1x^2$$

and $m_j(x) = 0$ for $j > 6$.

Simulation design (IV)

- ▶ Sparsity index $s = 11$ out of $pL = 3000$ total parameters.
- ▶ Signal-to-noise ratio:

$$\text{SNR} = \frac{\text{Var}(\sum_{j=1}^6 m_j(X_{it,j}))}{\text{Var}(F_t^\top \gamma_i + \varepsilon_{it})} \approx 1.$$

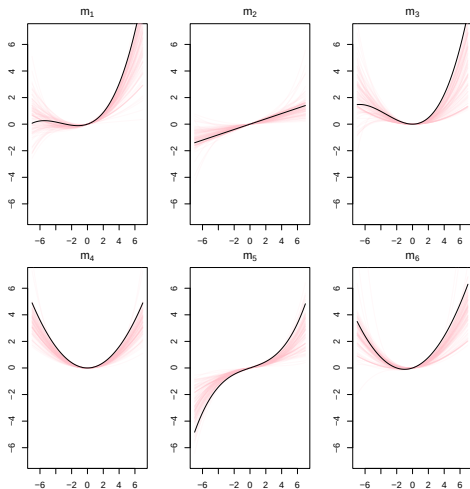
- ▶ Main question:
Can the unknown components $\beta_{j\ell}$ be recovered from the data?

Simulation design: estimators

Estimator	Label	Description
$\hat{\beta}_{\lambda}$	L	centering + $\hat{\Pi}$ + lasso
$\hat{\beta}_{\lambda}^{\text{oracle}}$	L-O	centering + oracle Π + lasso
$\hat{\beta}_{\lambda}^{\text{naive}}$	N	no centering + no projection + lasso

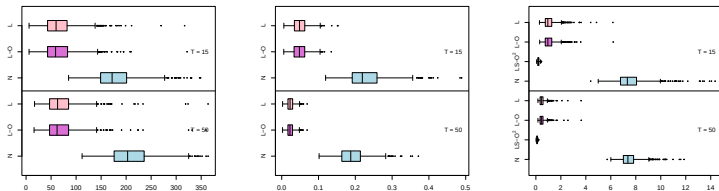
- ▶ L is the feasible HD-CCE estimator.
- ▶ L-O isolates the effect of estimating the projection matrix.
- ▶ N shows what happens if the factor structure is ignored.

Estimated component functions



Pink lines: $N = 100$ realizations $\hat{m}_j(x) = \sum_{\ell=1}^5 x^\ell \hat{\beta}_{\lambda,j\ell}$. Black line: True functions m_j .

- HD-CCE recovers the nonlinear component functions well.

Variable selection and ℓ_1 -error

a) Number of FP; b) Maximum size of FP; c) ℓ_1 -error norm
 Pink: HD-CCE; Violet: Oracle Lasso; Blue: Naive Lasso.

- ▶ Naive lasso produces more and larger false positives.
- ▶ HD-CCE has substantially smaller ℓ_1 -error rate.
- ▶ Oracle and feasible HD-CCE are close.

Takeaways

- ▶ We extend high-dimensional CCE estimation beyond the linear model.
- ▶ The estimator is simple to implement:
 - specify dictionary $\phi_{j\ell}$
 - construct projection matrix $\hat{\Pi}$
 - remove the factors with projection $\hat{\Pi}$
 - lasso on projected data

Takeaways

- ▶ We extend high-dimensional CCE estimation beyond the linear model.
- ▶ The estimator is simple to implement:
 - specify dictionary ϕ_{je}
 - construct projection matrix $\hat{\Pi}$
 - remove the factors with projection $\hat{\Pi}$
 - lasso on projected data
- ▶ Comprehensive Monte-Carlo experiments
- ▶ R-package with ready-to-use HD-CCE estimator on *GitHub*.
- ▶ Inference methodology $\rightarrow$ Test $H_0 : m_j = 0$ against $H_1 : m_j \neq 0$.
- ▶ Empirical application to the *Factor Zoo*: Investigate monthly stock returns and driving firm characteristics.

Literature



Bickel, Ritov & Tsybakov (2009). Simultaneous analysis of Lasso and Dantzig selector. *Annals of Statistics*.



Green, Hand & Zhang (2017). The Characteristics that Provide Independent Information about Average U.S. Monthly Stock Returns. *The Review of Financial Studies*.



Hansen (2007). Asymptotic properties of a robust variance matrix estimator for panel data when T is large. *Journal of Econometrics*.



Pesaran (2006). Estimation and inference in large heterogeneous panels with a multifactor error structure. *Econometrica*.



Rücker, Vogt, Linton & Walsh (2025). Estimation and inference in high-dimensional panel data models with interactive fixed effects. *Quantitative Economics*.

Appendix: Empirical application (I)

- ▶ Revisit the financial application from Rücker et al. (2025). Closely related is Green et al. (2017).
- ▶ Monthly returns R_{it} of ($n = 29$) Dow Jones stocks in time period April 2017 to March 2022 ($T = 60$).
- ▶ ($p = 90$) Firm characteristics $C_{i,t-1}$ observed at time $t - 1$.

Appendix: Empirical application (I)

- ▶ Revisit the financial application from Rücker et al. (2025). Closely related is Green et al. (2017).
- ▶ **Monthly returns** R_{it} of ($n = 29$) Dow Jones stocks in time period April 2017 to March 2022 ($T = 60$).
- ▶ ($p = 90$) **Firm characteristics** $C_{i,t-1}$ observed at time $t - 1$.
- ▶ Model:

$$R_{it} = \sum_{j=1}^p \sum_{\ell=1}^3 \phi_{\ell}(C_{i,t-1,j}) \beta_{j\ell} + \theta^{\top} D_{i,t-1} + \sum_{k=0}^K \gamma_{i,k} F_{t,k} + \varepsilon_{it}.$$

- ▶ Polynomial main effects and all pairwise interactions:

$$\phi_{\ell}(x) = x^{\ell}, \ell = 1, 2, 3 \quad D_{i,t-1} = \{C_{i,t-1,j} C_{i,t-1,j'} : j < j'\}.$$

- ▶ **High-dimensional setting:** $3p + \frac{p(p-1)}{2} = 270 + 4005 = 4275$ parameters, while $nT = 1740$.

Appendix: Empirical application (II)

- ▶ The HD-CCE estimator selects a sparse set of predictors.

Effect type	Candidate terms	Selected terms	Share selected
Main effects $\beta_{j\ell}$	270	4	1.48%
Interactions θ_h	4005	17	0.42%
Total	4275	21	0.49%

- ▶ The selected interaction terms suggest that characteristic effects are not purely linear.

Appendix: Empirical Application (III)

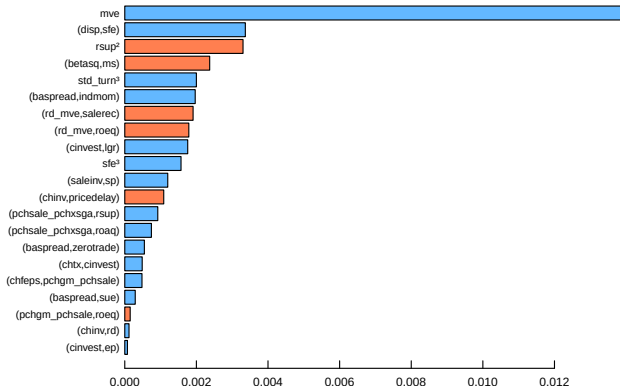
Table: Ticker symbols and company names of stocks used in the application.

Ticker	Company	Ticker	Company
<i>AAPL</i>	Apple Inc.	<i>INTC</i>	Intel Corporation
<i>AMGN</i>	Amgen Inc.	<i>JNJ</i>	Johnson & Johnson
<i>AMZN</i>	Amazon.com, Inc.	<i>JPM</i>	JPMorgan Chase & Co.
<i>AXP</i>	American Express Company	<i>KO</i>	The Coca-Cola Company
<i>BA</i>	The Boeing Company	<i>MCD</i>	McDonald's Corporation
<i>CAT</i>	Caterpillar Inc.	<i>MMM</i>	3M Company
<i>CRM</i>	Salesforce, Inc.	<i>MSFT</i>	Microsoft Corporation
<i>CSCO</i>	Cisco Systems, Inc.	<i>NKE</i>	Nike, Inc.
<i>CVX</i>	Chevron Corporation	<i>PFE</i>	Pfizer Inc.
<i>DIS</i>	The Walt Disney Company	<i>PG</i>	The Procter & Gamble Company
<i>GE</i>	General Electric Company	<i>TRV</i>	The Travelers Companies, Inc.
<i>GS</i>	The Goldman Sachs Group, Inc.	<i>UNH</i>	UnitedHealth Group Incorporated
<i>HD</i>	The Home Depot, Inc.	<i>VZ</i>	Verizon Communications Inc.
<i>HON</i>	Honeywell International Inc.	<i>WMT</i>	Walmart Inc.
<i>IBM</i>	Internat. Business Machines Corp.		

Appendix: Empirical Application (IV)

Acronym	Description	Acronym	Description
<i>absacc</i>	absolute accruals	<i>ms</i>	financial performance score
<i>acc</i>	working capital accruals	<i>mve</i>	size
<i>aeavol</i>	abnormal earnings announcement vol.	<i>mve-ia</i>	size (ind adj)
<i>age</i>	years since first Compustat coverage	<i>nanalyst</i>	analyst coverage
<i>agr</i>	asset growth	<i>nincr</i>	length of earnings run
<i>baspread</i>	bid-ask spread	<i>operprof</i>	operating profits / book equity
<i>beta</i>	beta	<i>orgcap</i>	organizational capital
<i>betasq</i>	beta squared	<i>pchcapx-ia</i>	% change in capital exp. (ind adj)
<i>bm</i>	book-to-market	<i>pchcurrat</i>	% change in current ratio
<i>bm-ia</i>	book-to-market (ind adj)	<i>pchdepr</i>	% change in depreciation to PP&E
<i>cash</i>	cash to assets	<i>pchgm-pchsale</i>	% change in sales
<i>cashdebt</i>	cash flow to debt	<i>pchquick</i>	% change in quick ratio
<i>cashpr</i>	cash productivity	<i>pchsale-pchinv</i>	% change in inventory
<i>cfp</i>	operating cash flow to price	<i>pchsale-pchrect</i>	% change in receivables
<i>cfp-ia</i>	operating cash flow to price (ind adj)	<i>pchsale-pchxsga</i>	% change in sales growth – % change in overheads
<i>chatoia</i>	change in asset turnover (ind adj)	<i>pchsaleinv</i>	% change in sales-to-inventory
<i>chcsho</i>	change in outstanding shares	<i>pctacc</i>	percent accruals
<i>chempia</i>	change in employees (ind adj)	<i>pricedelay</i>	price delay
<i>chfeps</i>	change in EPS forecast	<i>ps</i>	financial health score
<i>chinv</i>	change in inventory	<i>quick</i>	quick ratio
<i>chnanalyst</i>	change in analyst coverage	<i>rd</i>	large R&D increase
<i>chpmia</i>	change in profit margin (ind adj)	<i>rd-mve</i>	R&D to size
<i>cht</i>	change in taxes	<i>rd-sale</i>	R&D to sales
<i>cinvest</i>	corporate investment	<i>realestate</i>	real estate holdings
<i>convind</i>	convertible debt indicator	<i>roaq</i>	return on assets
<i>currat</i>	current ratio	<i>roavol</i>	earnings volatility
<i>depr</i>	depreciation to PP&E	<i>roeq</i>	return on equity
<i>disp</i>	EPS forecast dispersion	<i>roic</i>	return on invested capital
<i>divo</i>	dividend omission	<i>rsup</i>	revenue surprise
<i>dolvol</i>	dollar volume	<i>salecash</i>	sales to cash
<i>dy</i>	dividend yield	<i>saleinv</i>	sales to inventory
<i>egr</i>	growth in book value of equity	<i>salerec</i>	sales to receivables
<i>ep</i>	earning to price	<i>secured</i>	secured debt
<i>egr5yr</i>	5yr EPS growth forecast	<i>securedind</i>	secured debt indicator
<i>gma</i>	gross profits to assets	<i>sfe</i>	scaled earnings forecast
<i>grcapx</i>	growth in capital expenditure	<i>sgr</i>	sales growth
<i>grltnoa</i>	growth in long term net-op. assets	<i>sp</i>	sales to price
<i>herf</i>	industry sales concentration	<i>std-dolvol</i>	volume volatility
<i>hire</i>	employment growth	<i>std-turn</i>	turnover volatility
<i>idivol</i>	idiosyncratic return volatility	<i>stdacc</i>	accrual volatility
<i>indmom</i>	industry momentum	<i>stdcf</i>	cash flow volatility
<i>invest</i>	capital expenditures and inventory	<i>sue</i>	surprise earnings
<i>lev</i>	leverage	<i>tang</i>	asset tangibility
<i>lgr</i>	growth in liabilities	<i>tb</i>	tax to book income
		<i>turn</i>	turnover
		<i>zerotrade</i>	(weighted) days with zero trades

Appendix: Empirical Application (V)



- ▶ Only four nonlinear/main-effect terms are selected: mve, rsup², sfe³, and std_turn³.
- ▶ The remaining selected terms are interactions, suggesting relevant nonlinear cross-characteristic effects.